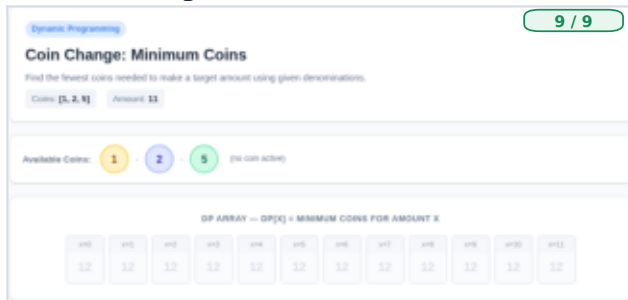


Cross-family gallery: four selected AlgoTutorGen tutors

(a) Coin Change · DP



Dynamic Programming

Coin Change: Minimum Coins

Find the fewest coins needed to make a target amount using given denominations.

Coins: [1, 2, 5] Amount: 13

Available Coins: 1 2 5 (no coin active)

DP ARRAY — $DP[X]$ = MINIMUM COINS FOR AMOUNT X

amt	amt	amt	amt	amt	amt	amt	amt	amt	amt	amt	amt	amt
1	2	3	4	5	6	7	8	9	10	11	12	13
1	2	2	2	2	3	3	3	3	4	4	4	4

(b) Dijkstra · Shortest path



Emergency Dispatch — Dijkstra's Algorithm

Verified Result 9 / 9

Network Graph

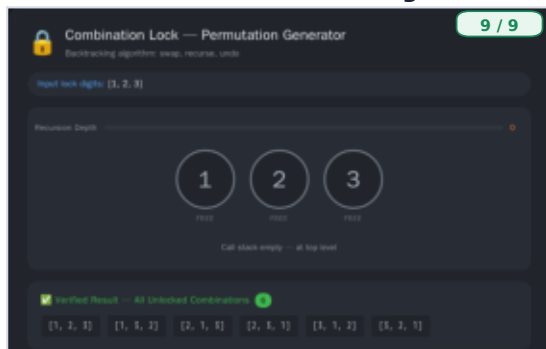
Nodes: A, B, C, D

Edges: A-B, A-C, A-D, B-C, B-D, C-D

Build an internal directed graph representation from weighted_graph input. We need to efficiently access neighbors and edge weights during relaxation.

Step 8 / 17

(c) Permutations · Backtracking



Combination Lock — Permutation Generator

Backtracking algorithm: swap, recurse, undo

Input lock digits: [1, 2, 3]

Recursion Depth

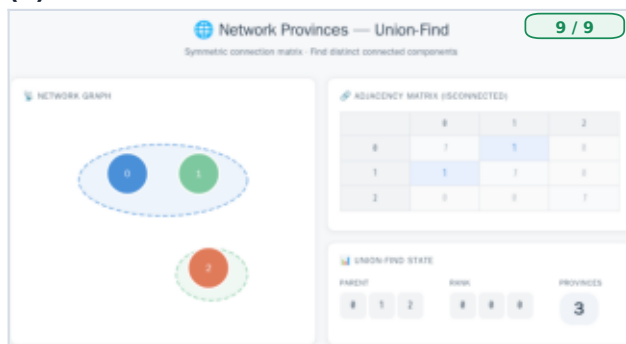
1 2 3

Call stack empty — at top level

Verified Result — All Unlocked Combinations

[1, 2, 3] [1, 3, 2] [2, 1, 3] [2, 3, 1] [3, 1, 2] [3, 2, 1]

(d) Provinces · Union-find



Network Provinces — Union-Find

Symmetric connection matrix. Find distinct connected components.

Network Graph

Agency Matrix (Disconnected)

	0	1	2
0	1	1	0
1	1	1	0
2	0	0	1

Union-Find State

PARENT: 0 1 2 RANK: 0 0 0 PROVINCES: 3